An Autonomous Multi-Agent System for Astronomy Literature Fact-Checking

Research Proposal

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Abstract

This project develops an autonomous multi-agent system designed to bridge the gap between unstructured astronomical literature and structured celestial catalogs. Leveraging domain-specific Natural Language Processing (NLP) models like AstroBERT and Large Language Models (LLMs), the system identifies celestial objects and their reported physical parameters, such as mass, temperature, and coordinates, within full-text research articles. To ensure scientific integrity, the framework employs an orchestration layer where specialized agents programmatically query professional astronomical databases including SIMBAD, Gaia, and NED through the Astroquery API to cross-verify findings against established ground truth data. The resulting agentic summary synthesizes the paper's contributions while highlighting potential data conflicts or novel discoveries, offering a scalable solution for managing the massive influx of information in modern astrophysics.

1. Research Context and Problem Statement

There has been a phenomenal surge in the amount of astronomy literature over the last century. Modern observatories generate more data in a single night than older ones used to in decades (Garraffo, 2024). This has significantly spiked astronomy literature publication rates, as more and more researchers write about new satellite missions' discoveries. While there is a wealth of structured astronomy databases, a large number of these discoveries remain trapped in the free text of journals, making it difficult to systematically track them (Gough, 2024). This growing volume of data makes manual verification an increasingly tedious and impractical task (Chen et al., 2022). The issue is further exacerbated by the findings of a 2014 study by Pepe et al. (2014) on papers published in American Astronomical Society journals in 2001. The researchers found that 44% of the data links in those papers were broken by 2011, making it difficult to fact-check them. These findings underscore the need for a system that can extract celestial objects and their physical parameters from free-text literature, cross-verify them against official catalogs, flag any conflicts, and highlight new data.

Although no one has fully addressed this need in its entirety, various researchers have separately tackled its different aspects. In 2021, Grezes et al. (2021) developed astroBERT, a domain-specific language model tailored to astronomy literature and highly performant on named entity recognition (NER). While astroBERT is a powerful tool for the task at hand — particularly given that it was pre-trained on about 395,000 astronomy papers — its NER performance was evaluated only for astronomy organization names in acknowledgment sections, not for celestial objects or physical parameters appearing elsewhere in papers. As Dai and Karimi (2022) report, this limitation reflects a broader challenge in the field: lexical ambiguity makes named entity extraction in astronomy literature especially difficult, as the same term can refer simultaneously to a celestial object, a scientist, an institution, or a mission.

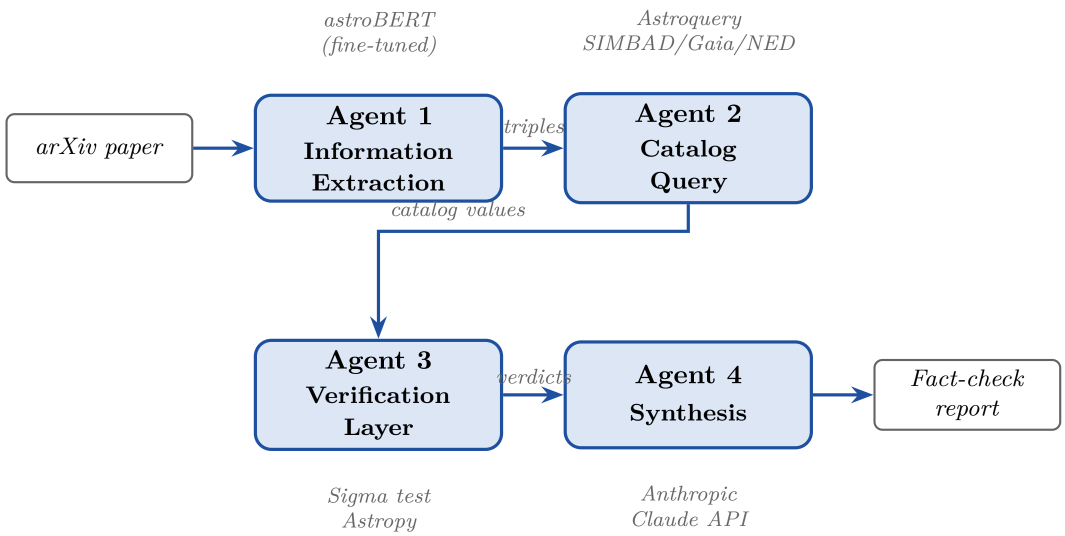
While Shapurian et al. (2023) have addressed this difficulty in the context of planetary surface features, their work does not address the fact-checking component of our problem, and it is limited to a specific named entity category. Recognizing that NER alone is insufficient to handle lexical ambiguity, they employ a multi-step pipeline combining rule-based filtering, part-of-speech tagging, knowledge graph matching, and LLM-based contextual reasoning. For cross-verification against official databases, tools such as C3 (Riccio et al., 2016) enable cross-matching across large astronomical catalogs, but operate on structured inputs and have no capability to handle unstructured text.

On the infrastructure side, Gemelli et al. (2024) argue that approaches relying on a single model are insufficient for tasks involving complex scientific knowledge management. Trinh et al. (2025) suggest that an autonomous multi-agent system is better suited to fact-checking complex information. Their system comprises an input ingestion agent, a query generation agent, an evidence retrieval agent, and a verdict prediction agent. It is, however, not tailored to verifying reported data about celestial bodies, having been originally developed for fact-checking general online content. The ADS infrastructure has itself evolved significantly, with NLP and machine learning techniques now being applied to metadata enrichment and semantic search (Accomazzi, 2024).

The specific research problems we aim to solve are: (1) whether an autonomous multi-agent system can reliably extract physical parameters from astrophysical literature, and (2) whether comparison against catalog data can highlight meaningful conflicts at a speed that manual review cannot match.

2. Proposed Solution

To address the difficulties of designing an astronomy literature fact-checking system in light of the insights and limitations of previous research mentioned above, we propose a multi-agent pipeline with four agents coordinated with LangGraph. Our first agent extracts celestial body names, physical parameter names, and their reported values. Our second agent queries official astronomy databases based on the type of celestial body identified. Our third agent compares the extracted values against canonical data, flagging discrepancies, highlighting data absent from any official catalog, and confirming data that conforms with official records. Our fourth agent summarizes the results, producing a human-readable synthesis of conflicting, valid, novel, and manually-verifiable data.



2.1. Agent 1: Information Extraction

For information extraction, we use astroBERT, a language model created by Grezes et al. (2021) and trained on approximately 395,000 astronomy research papers. This first agent reads the input research paper and extracts, for every celestial body mentioned, the entity's name, its physical parameters, and the parameters' reported numerical values. Using span-based Named Entity Recognition (NER) — which has been shown to be more effective than word-level NER (Dai and Karimi, 2022) — astroBERT forms triples for each named entity, matching the celestial body to its parameter type and numerical value.

Before deploying astroBERT for this task, fine-tuning is necessary. As Grezes et al. (2021) note, the model's NER was originally trained to extract organization names from acknowledgement sections, not astronomical objects and their numerical parameters.

Recent work motivates our fine-tuning strategy. Santoso et al. (2024) demonstrated the potential of fine-tuning language models with a small set of human-annotated samples to generate labeled NER training data. Shao (2025) showed that prompting a pre-trained language model with domain-specific information for celestial object identification improves its ability to recognize astronomy-specific entity formats.

Building on these insights, we begin with 50 to 100 human-annotated astronomy sentences drawn from arXiv papers, each marking entity spans — object names, parameter types such as orbital period, stellar mass, and redshift, and numerical values with units — using the open-source annotation tool Label Studio. We then prompt an LLM (Claude 3 Sonnet, OpenAI o3-mini, or similar) to act as a domain expert annotator, providing it with astronomy-specific formatting conventions and the human-labeled samples as reference, and instructing it to generate new, realistic astronomy sentences with labeled entities in the same format. Repeating this process with different seed samples scales our dataset from 100 human-labeled sentences to over 1,000 LLM-generated labeled sentences, with a human verifying a random selection of the generated material.

To give the model multi-sentence context during training, we apply a sliding window that feeds overlapping groups of three to four sentences at a time. This provides sufficient context for references spanning multiple sentence boundaries without requiring the model to process entire papers at once, which would likely exceed its computational capacity.

Following Dagdelen et al. (2024)'s approach to structured information extraction from scientific texts, we fine-tune astroBERT as a sequence-to-sequence (seq2seq) model to read labeled sentences and output a structured triple for each named entity. A lightweight parsing step then converts these triples into the `(object, param_type, value)` dictionaries that Agent 3 expects.

2.2. Agent 2: Catalog Query

For catalog querying, we use Astroquery, a Python-based toolkit for searching online astronomical databases. For each triple produced by Agent 1, this agent searches the celestial object's name, parameter type, and numerical value across three premier databases: SIMBAD, Gaia, and NED.

The agent first queries the Set of Identifications, Measurements, and Bibliography for Astronomical Data (SIMBAD). SIMBAD contains bibliographic information, object classifications, coordinates, magnitudes, proper motions, parallax, velocity, redshift, size, spectral and morphological types, and cross-identifications for a wide range of astronomical objects (Centre de Données astronomiques de Strasbourg, 2024). We prioritize SIMBAD because its cross-identification feature helps resolve naming ambiguities. Moreover, even when SIMBAD lacks specific parameter data for an object, it typically records the object's identifiers and nature — confirming that the object is officially recognized and informing the agent's next query choice.

If the object is a star, the agent queries the European Space Agency's Gaia DR3, a catalog containing positions, distances, compositions, brightnesses, and velocities for nearly two billion stars, primarily within the Milky Way (European Space Agency, 2024). Gaia queries use a TAP (Table Access Protocol) interface, a standard structured query language for astronomical databases. If the object is extragalactic, the agent instead queries the NASA/IPAC Extragalactic Database (NED), which specializes in galaxies and objects beyond the Milky Way (NASA/IPAC Extragalactic Database, 2024).

The resulting priority chain is: SIMBAD first, Gaia DR3 second for stellar parameters, and NED second for extragalactic parameters. If all three catalogs are exhausted without a match, the agent returns a `not_found` status. An unresolvable object may represent a new discovery, a recently named object not yet cataloged, or a non-standard name used by the paper's authors.

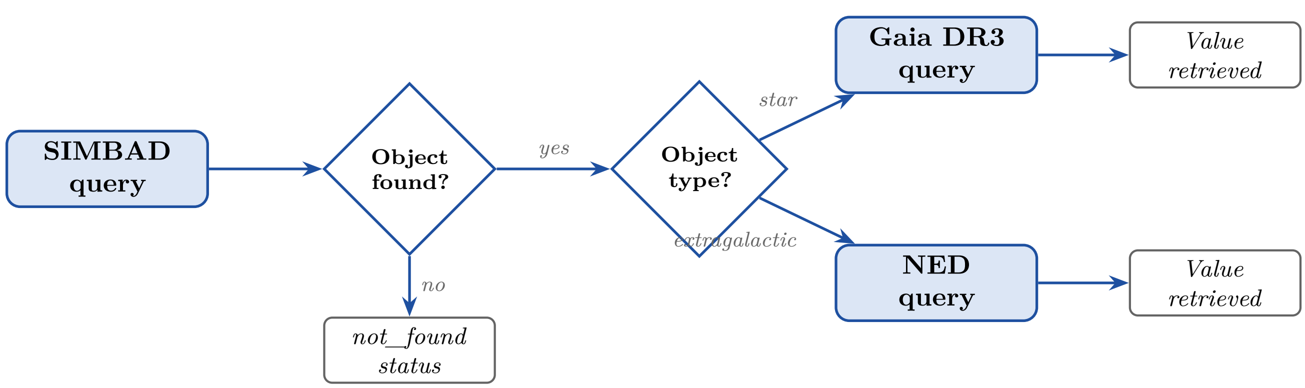


Figure 1: Agent 2 catalog query priority chain. Every object is first submitted to SIMBAD for identification and alias resolution. Stellar objects then go to Gaia DR3; extragalactic objects go to NED. Objects not found in any catalog receive a `not_found` status for Agent 3.

2.3. Agent 3: Verification Layer

The third agent performs the actual scientific fact-checking, with Agents 1 and 2 preparing its inputs and Agent 4 communicating its outputs. Implemented in LangGraph, it receives a value from the input paper and a corresponding value from a catalog, then determines whether the two are consistent, discrepant, strongly discrepant, or potentially novel.

To make principled, quantitative judgments, the agent applies a sigma test: it computes the absolute difference between the two values and divides it by their combined uncertainty. If both the paper and the catalog report an uncertainty, the agent computes the combined uncertainty as the square root of the sum of their squares; if only the catalog reports an uncertainty, that value is used alone. The resulting σ value determines the verdict:

- **Consistent** ($\sigma < 2$): the paper's value and the catalog value agree within their combined uncertainties. This is the expected outcome for most well-measured, well-cataloged parameters.
- **Discrepant** ($2 \leq \sigma \leq 5$): the values differ beyond what measurement uncertainty alone can explain. This does not necessarily imply an error, but warrants human verification.
- **Strongly discrepant** ($\sigma > 5$): a serious conflict that almost certainly cannot be attributed to measurement error, and requires immediate human review.
- **Potentially novel**: either the object was found in a catalog but the specific parameter has no recorded value, or the object itself could not be found in any catalog. The former suggests the paper is adding new measurements; the latter suggests the paper may be reporting an object not yet cataloged, which could represent a genuine discovery.

Table 1: Agent 3 verdict categories and their sigma thresholds.

Verdict	**Condition**	**Interpretation**
Consistent	$\sigma < 2$	Values agree within measurement uncertainty
Discrepant	$2 \leq \sigma \leq 5$	Difference exceeds uncertainty; warrants review
Strongly discrepant	$\sigma > 5$	Serious conflict; requires immediate review
Potentially novel	No catalog value	Paper may be reporting new data or new object

The agent outputs a structured dictionary that Agent 4 uses to generate its human-readable report. Before running the sigma test, Agent 3 performs unit conversion, since the paper and the catalog may report the same parameter in different units. The `astropy.units` submodule handles this automatically.

2.4. Agent 4: Synthesis

The fourth agent synthesizes the full pipeline's output into a human-readable report. We implement it using the Anthropic Claude API. A key risk with LLM-based summarization is hallucination — the generation of plausible-sounding content not grounded in the model's input. To mitigate this, we draw on Ji et al. (2023)'s finding that structured inputs, explicit output constraints, and reduced generation randomness substantially lower hallucination rates in summarization tasks.

We design the synthesis agent around the principle of grounded generation: the model is instructed to produce text that describes only what the input dictionary explicitly contains, and is explicitly forbidden from drawing on its pre-training knowledge to supplement the summary. The prompt specifies the exact output structure, requires the model to cite the sigma value and verdict category for every triple, and instructs it to flag any triple for which a parameter was not found in any catalog rather than silently omitting it.

3. Evaluation and Implementation Plan

3.1. Evaluation Process

We will evaluate each agent individually, then run a comprehensive evaluation. This makes the debugging phase easier, allowing us to spot potential problems early on at each phase.

Evaluating the Information Extraction Layer

We evaluate the first agent using a set of around 150 annotated sentences from the annotation phase, making sure to select sentences that the model has not seen before. The standard evaluation split is 70/15/15 for train/validation/test. Fifteen percent of 1,000 (total number of sentences we are using) is 150. We use a set of three metrics to assess Agent 1's performance: precision, recall, and F1. Precision evaluates the fraction of spans the model got right of all the spans it predicted while recall assesses the portion the model found of all the spans that exist. F1 is the harmonic mean of the two, and is the standard metric for NER tasks.

We count a triple as correctly extracted by Agent 1 only if all three components are correctly identified, meaning the right object name, parameter label, and numerical value at the same time. We compute the three metrics for each triple of the 150-sentence test set.

Dai and Karimi (2022) used an F1 threshold of 0.80 for their NER task. But given that our NER task is more complex, we lower our F1 threshold to 0.75. If the agent falls below threshold, we will generate more data and repeat the fine-tuning process.

Evaluating the Catalog Query Agent

Agent 2 takes an object's name from a triple from Agent 1 and looks for the right astronomical catalog to find the equivalent canonical value. We evaluate Agent 2's ability to retrieve the right value from the appropriate database.

Our evaluation set is made of 70 object–parameter pairs pulled from arXiv papers. We select pairs for which we know the right catalog result, which allows us to check the agent's output against a reliable ground truth (Chen et al., 2022). Following best practice in retrieval evaluation, we ensure our benchmark set is diverse, with 20 pairs from SIMBAD, 20 from Gaia, 20 from NED, and 5 random aliases that cannot be found in any databases, and 5 real but non-popular aliases, helping to assess the agent's ability to handle alias resolution with uncommon object names (Wenger et al., 2000). The goal of this split is to ensure the set reflects the full scope of the range of pairs the system will face.

For each pair, we check if the agent returned the correct value from the right database and compute the success rate. We also record the specific reason for every failure as we expect failures for when the agent cannot find the object's name in any catalogue, or the object is found in the catalog but not the parameter, or the query itself is constructed incorrectly (Dai and Karimi, 2022). This separation informs the appropriate solution for each failure case.

Our target success rate is 80% because edge cases are inevitable. For example, objects might be recently named objects not yet indexed, or parameter fields that a catalog records for some objects but leaves empty for others (Lucey et al., 2025). This threshold leaves room for the hard alias-resolution and not-found cases to be tracked separately.

Evaluating the Verification Layer

For Agent 3, we constitute a test set of 60 paper–catalog value pairs, 15 examples from each of the four verdict categories: consistent, discrepant, strongly discrepant, and potentially novel. We label each pair independently and resolve any disagreements with our advisor. To measure each annotator's consistency with the verdict criteria, we compute Cohen's Kappa, the standard measure of agreement between two annotators (James, 2025). We consider a Kappa score above 0.80 as an indicator of strong agreement (James, 2025), and anything below as an invitation to revise the verdict guidelines. We evaluate Agent 3's overall accuracy by computing the fraction of pairs it correctly evaluated. We set a threshold of 85% for overall accuracy and 90% recall on the discrepant and strongly discrepant categories, since missing a serious conflict is a more consequential error than raising a false alarm (Trinh et al., 2025).

Evaluating the Synthesis Agent

This agent takes Agent 3's output, a dictionary of triples with a sigma value and a verdict, and produces a clean, human-readable summary report. Since we are using the Anthropic API to handle this task, the main concern at this stage is hallucination.

According to the hallucination literature, the most reliable method to detect LLM hallucination is human sentence-level annotation, which implies a person evaluating the reliability of each sentence output against real data (Ji et al., 2023). In accordance with the literature, we use 30 test inputs to assess Agent 4, made-up dictionaries akin to the output of Agent 3. We deliberately vary their content, including consistent triples, triples with flagged conflicts, and ones with objects not found in any catalog. This is because testing only inputs where everything is consistent would not let us know the model's propensity for filling the gaps (Ji et al., 2023). For each of the 30 outputs, both of us read through the report sentence by sentence and assess the model's output by computing the hallucination rate (number of sentences with made-up facts divided by total number of sentences). We aim for a hallucination rate below 5%, which gives us room for natural language flexibility while still holding the model to a tight standard of faithfulness.

Finally, we show a sample of reports to three to five people with astronomy backgrounds and ask them if the report gives a cogent summary of the paper's findings, and we integrate their feedback in our report design.

Evaluating the Entire Pipeline

We will run the full LangGraph pipeline on 10 complete arXiv papers, making sure the papers contain a variety of different celestial body types (stellar, extragalactic, and solar system) as well as different parameter types.

Before running the pipeline, we both annotate each paper and evaluate each triple against official catalogs. We then run the pipeline and compare its results with ours and evaluate two metrics: precision and recall. Precision evaluates the fraction the pipeline got right out of every triple it extracted. This tells us how much we can trust the pipeline. A low precision indicates that the report is full of errors, and is therefore unreliable. Recall evaluates the fraction of triples the pipeline detected of all the triples in the paper. This tells us how good the pipeline is at extracting the data from the papers.

We also time the full evaluation cycle for each paper to assess the model's efficiency. We compare this with the time it took us to annotate and fact-check the same number of papers. With all these tests, we try to answer these questions: if a scientist read only our pipeline's report without manually fact-checking its results, would they get an accurate picture of the truth? Would the serious conflicts be clearly visible? Would the confirmed findings be clearly confirmed? Would anything important be buried or missing? Does this pipeline fare better than manual fact-checking?

Table 2: Summary of evaluation targets for each agent and the end-to-end pipeline.

Component	**Metric**	**Target**
Agent 1	Span-level F1	≥ 0.75
Agent 2	Success rate	$\geq 80\%$ on well-known objects
Agent 3	Overall accuracy	$\geq 85\%$
Agent 3	Recall (discrepant categories)	$\geq 90\%$
Agent 4	Hallucination rate	$< 5\%$
End-to-end	Precision and recall	Both reported
End-to-end	Runtime per paper	$<$ 10 minutes

3.2. Timeline

We have tried to be realistic here. The annotation and fine-tuning stages are the ones we are least certain about, so we front-loaded them and built in a buffer week near the end. Poster prep does not start until the evaluation is done, so we are not putting ourselves in a position where we are still running experiments while trying to make a poster.

Table 3: Project timeline across 14 weeks.

Week	Task	Est. Hours	Key Tools
1	Setup and annotation guidelines	10 hrs	Label Studio
2	Seed annotation (100 sentences)	15 hrs	Label Studio
3–4	LLM data augmentation (1,000 sentences)	20 hrs	Claude API, scikit-learn
5–6	Fine-tune and evaluate Agent 1	25 hrs	Hugging Face, seqeval
7	Build and evaluate Agent 2	15 hrs	Astroquery
8	Build and evaluate Agent 3	12 hrs	Astropy
9	Build and evaluate Agent 4	12 hrs	Anthropic SDK
10	End-to-end pipeline evaluation	15 hrs	LangGraph
11	Buffer week	10 hrs	—
12–13	Poster preparation	20 hrs	—
14	Final revisions and submission	8 hrs	—
	Total	**162 hrs**	

Week 1 (approximately 10 hours). We set up our annotation environment and write our annotation guidelines. We install Label Studio locally following the official Label Studio documentation at labelstud.io and configure it for span-level NER with three label types: OBJECT, PARAM TYPE, and VALUE. We each read at least two arXiv papers from our target domain to calibrate what kinds of sentences we will be annotating before we start. The goal by the end of this week is to have a working Label Studio setup and a written annotation guide both of us have agreed on.

Week 2 (approximately 15 hours). We each independently annotate 50 arXiv sentences in Label Studio, then compare our labels and reconcile disagreements. We compute Cohen's Kappa between our two annotation sets, following the procedure described in James (2025). If Kappa falls below 0.80, we revise the guidelines and re-annotate the disputed sentences before moving on. By the end of this week we should have approximately 100 clean, agreed-upon labeled sentences ready to serve as seed data.

Weeks 3 and 4 (approximately 20 hours). We scale the dataset from 100 human-labeled sentences to over 1,000 labeled sentences using LLM-augmented generation. We prompt Claude 3 Sonnet or OpenAI o3-mini via their respective APIs, providing the human-labeled sentences as examples and instructing the model to generate new realistic astronomy sentences with labeled entities in the same format, following the approach described in Santoso et al. (2024). We manually verify a random 10% sample of the generated sentences to make sure the quality is there, then split everything into training, validation, and test sets at a 70/15/15 ratio using scikit-learn's train test split function, which is documented at scikit-learn.org.

Weeks 5 and 6 (approximately 25 hours). We fine-tune astroBERT on our labeled dataset. We load the model from Hugging Face at adsabs/astroBERT using the Hugging Face Transformers library, documented at huggingface.co/docs/transformers. We follow the seq2seq fine-tuning approach described in Dagdelen et al. (2024) and apply a sliding window of three to four sentences as described in our solution section. We use the Hugging Face Trainer API for the training loop and evaluate on the validation set after each epoch. Once training is done, we evaluate Agent 1 on the held-out test set using the seqeval library, documented at github.com/chakki-works/seqeval, computing span-level precision, recall, and F1 for each entity type as well as strict triple-level accuracy. If F1 falls below 0.75, we generate more training data and fine-tune again before moving forward. This is probably the step most likely to take longer than expected.

Week 7 (approximately 15 hours). We build and evaluate Agent 2. We implement the SIMBAD, Gaia DR3, and NED query chain using the Astroquery library, documented at astroquery.readthedocs.io. SIMBAD queries use Astroquery's built-in SIMBAD module. Gaia queries use the TAP interface via Astroquery's Gaia module. NED queries use Astroquery's IPAC Ned module. We run Agent 2 on our 70-pair stratified benchmark and record success rates and failure modes as described in the evaluation plan. We expect some messiness around object naming and not-found cases, so we are giving this its own week.

Week 8 (approximately 12 hours). We build and evaluate Agent 3. We implement the sigma test using Python arithmetic and handle unit conversion with the `astropy.units` submodule, documented at docs.astropy.org/en/stable/units. We construct our 60-pair benchmark by pulling paper-catalog value pairs from published correction notices and catalog updates as described in the evaluation plan. We compute overall accuracy and per-category recall and adjust the sigma thresholds if recall on the discrepant categories falls below 90%.

Week 9 (approximately 12 hours). We build and evaluate Agent 4. We implement it using the Anthropic Python SDK, documented at docs.anthropic.com. We finalize the grounded-generation prompt following Ji et al. (2023)'s recommendations on structured inputs and explicit output constraints. We run the hallucination evaluation on 30 hand-crafted input dictionaries and collect readability feedback from three to five astronomy volunteers as described in the evaluation plan. If the hallucination rate comes in above 5%, we revise the prompt before moving on.

Week 10 (approximately 15 hours). We wire all four agents together into the full LangGraph pipeline. LangGraph is documented at langchain-ai.github.io/langgraph. We run the end-to-end evaluation on 10 complete arXiv papers as described in the evaluation plan, recording precision, recall, and runtime per paper. We note any failure patterns that still need addressing.

Week 11 (approximately 10 hours). Buffer week for whatever issues come up in the end-to-end run.

Weeks 12 and 13 (approximately 20 hours). We make the poster. We put together the visualizations, write the narrative sections, and get feedback from our advisor before the final version.

Week 14 (approximately 8 hours). Final revisions and submission. We also do a short retrospective on what worked, what did not, and what we would approach differently if we started over.

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